Laboratory Course: [BCS-351] Data Structures Lab

Programme: Bachelor of Technology in Computer Science & Engineering **Manual Type: LAB MANUAL**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Department of Computer Sc**

Official Laboratory Manual: Data Structures Lab

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1. Laboratory Objectives & Evaluation Structure

- Course Goal: Practical mastery of linear and non-linear data structures in ANSI C/C++.
- Evaluation Breakdown: Continuous Assessment (20 Marks) + End-Semester Lab Exam (30 Marks).
- Rubrics: Algorithm design (20%), clean implementation (40%), viva voce (20%), lab record (20%).

2. Core Prescribed Experiments Schedule

- Week 1-3: Array insertion/deletion, Stack push/pop, Infix to Postfix expression conversion.
- Week 4-6: Linear Queue, Circular Queue using modulo arithmetic, and Priority Queue.
- Week 7-9: Singly Linked List, Doubly Linked List, and Polynomial Addition using Linked Lists.
- Week 10-12: Binary Search Tree (BST) insertion, search, and traversals (In-order, Pre-order, Post-order).
- Week 13-14: Sorting algorithms (Quick Sort, Merge Sort, Heap Sort) and Search comparisons.